evidence-computaion-from-overlap

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Derivation: Joint Evidence from Posterior Overlap

Likelihood factorization:

$$L_{\text{tot}}(\theta_s, \theta_{\text{EM}}, \theta_{\text{GW}}) = L_{\text{EM}}(d_{\text{EM}} \mid \theta_s, \theta_{\text{EM}})\, L_{\text{GW}}(d_{\text{GW}} \mid \theta_s, \theta_{\text{GW}}), \quad \pi(\theta) = \pi(\theta_s)\pi(\theta_{\text{EM}})\pi(\theta_{\text{GW}})$$

Step 1 — Marginal likelihood of θ_s from each dataset:

$$L_{\text{EM,marg}}(\theta_s) \equiv \int L_{\text{EM}}(d_{\text{EM}} \mid \theta_s, \theta_{\text{EM}})\, \pi(\theta_{\text{EM}})\, d\theta_{\text{EM}} \;\Rightarrow\; Z_{\text{EM}} = \int \pi(\theta_s)\, L_{\text{EM,marg}}(\theta_s)\, d\theta_s$$

(analogously for GW).

Step 2 — Bayes' theorem on the marginal problem:

$$p(\theta_s \mid d_{\text{EM}}) = \frac{\pi(\theta_s) L_{\text{EM,marg}}(\theta_s)}{Z_{\text{EM}}} \;\Rightarrow\; L_{\text{EM,marg}}(\theta_s) = \frac{Z_{\text{EM}}\, p(\theta_s \mid d_{\text{EM}})}{\pi(\theta_s)}$$

$$L_{\text{GW,marg}}(\theta_s) = \frac{Z_{\text{GW}}\, p(\theta_s \mid d_{\text{GW}})}{\pi(\theta_s)}$$

Step 3 — Joint evidence, integrate out nuisance params:

$$Z_{\text{joint}} = \int \pi(\theta_s)\, L_{\text{EM,marg}}(\theta_s)\, L_{\text{GW,marg}}(\theta_s)\, d\theta_s$$

Step 4 — Substitute Step 2, cancel one prior copy:

$$Z_{\text{joint}} = \int \pi(\theta_s) \cdot \frac{Z_{\text{EM}}\, p(\theta_s \mid d_{\text{EM}})}{\pi(\theta_s)} \cdot \frac{Z_{\text{GW}}\, p(\theta_s \mid d_{\text{GW}})}{\pi(\theta_s)}\, d\theta_s$$

$$Z_{\text{joint}} = Z_{\text{EM}}\, Z_{\text{GW}} \int \frac{p(\theta_s \mid d_{\text{EM}})\, p(\theta_s \mid d_{\text{GW}})}{\pi(\theta_s)}\, d\theta_s$$